

Christian pioneers: ⁸⁷Sr/⁸⁶Sr isotope analysis at Kata Gård an early Christian site in Varnhem, Sweden

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Background

- Kata Gård in Varnhem (Fig.1) offers an opportunity to study mobility during the Christianization period (9th–11th c. AD)
- Socially stratified cemetery with ~3000 burials
- Cellar was dated to the 11thc. AD, making it one of the earliest Christian churches in Sweden (Fig. 2)¹
- Region features diverse geology (⁸⁷Sr/⁸⁶Sr range = 0.70 to >0.74) (Fig. 3)



Fig. 1 Varnhem location (Green) in reference to other early Christian sites (Blue) and modern cities (Red)



Fig. 2 Church stone cellar

Methods and Materials

- Molar enamel from 25 individuals (8♂, 9♀, 8 subadults) were sampled
- Local ranges defined by 5 km and 30 km (Fig. 3)
- LA-MC-ICP-MS measured ⁸⁷Sr/⁸⁶Sr ratios (Fig. 4)

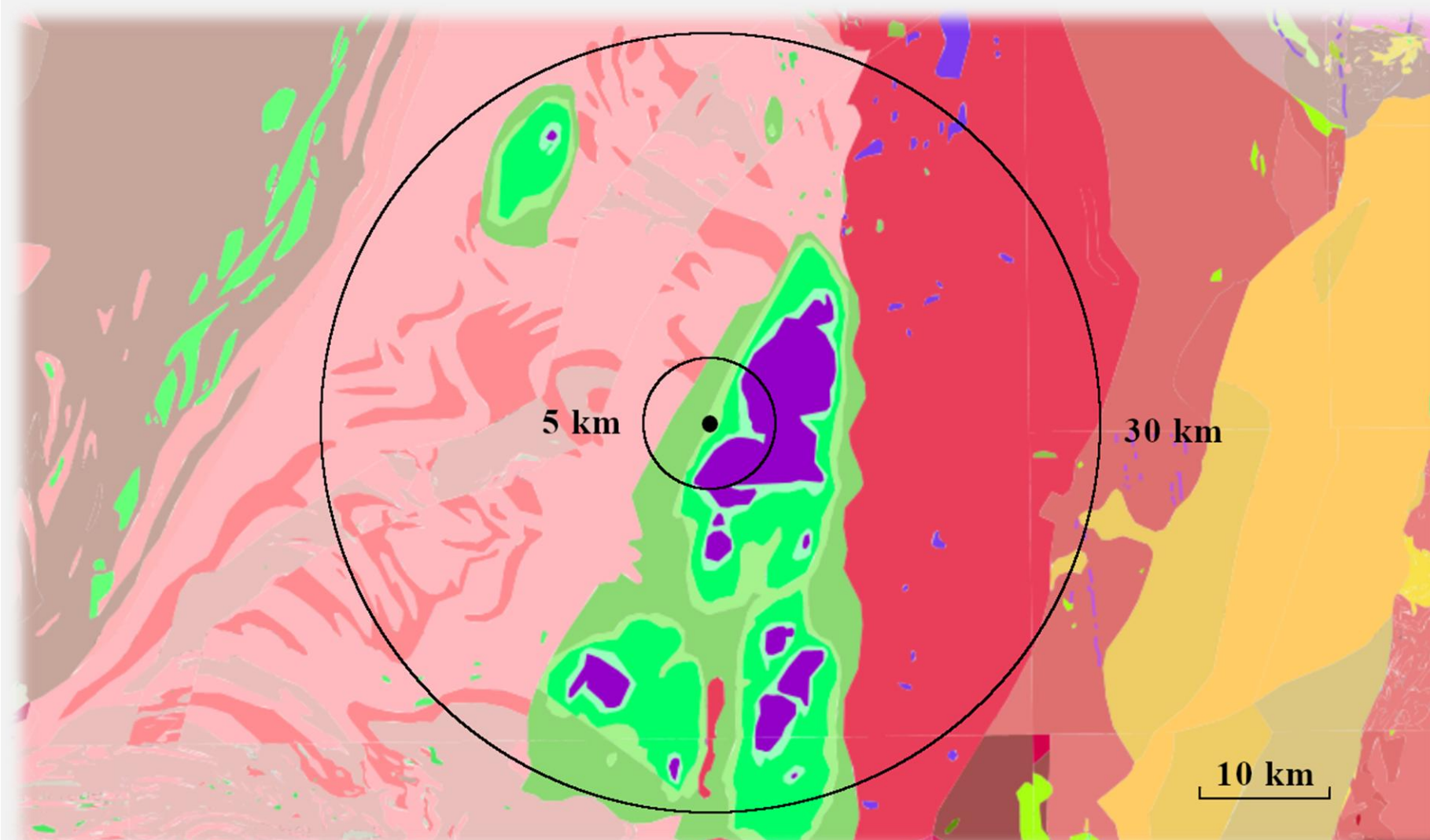
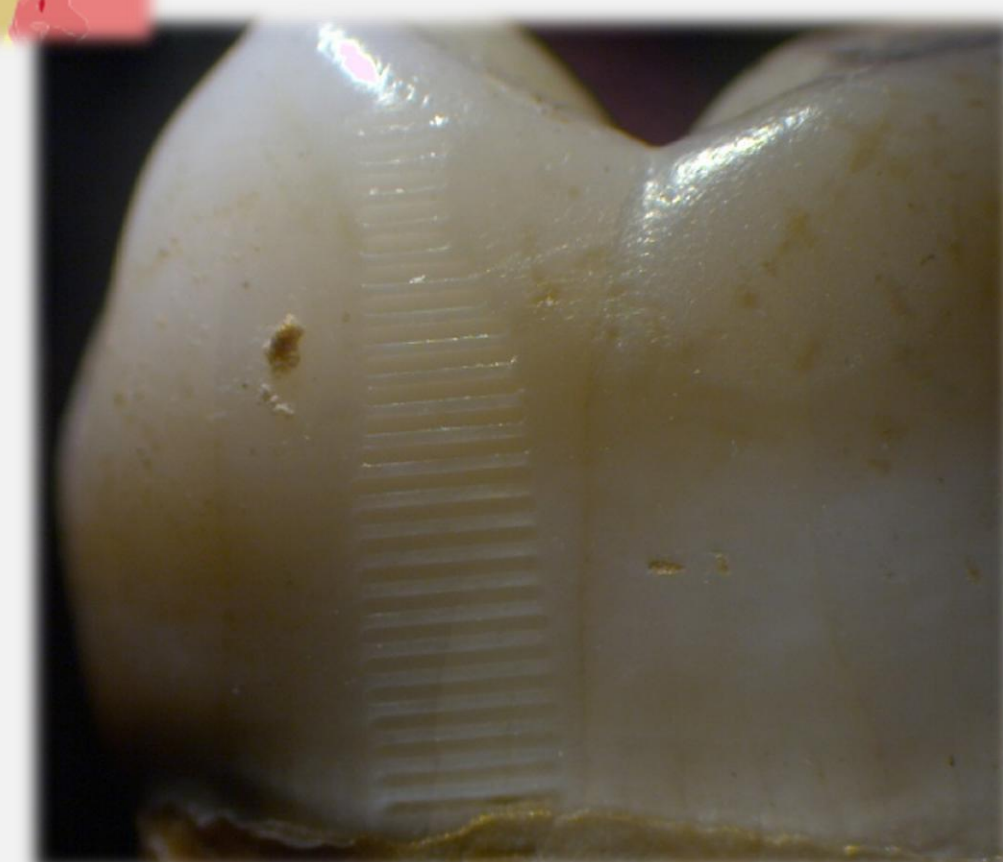


Fig. 3 Geological bedrock) variation 5-30 km and beyond of Varnhem (black dot). Taken from the geological maps by the Geological Survey of Sweden (SGU).

Fig. 4 Multiple ablation lines yield high-resolution mobility profiles. Photo is from Individual 110, values are shown in Fig. 5 under 'permanent enamel values' in green. The values indicate a local origin within 5 km.



Results

- Adult ablated lines were mostly nonlocal (86.6%) at 5 km and local (60.8%) at 30 km (Fig. 5)
- Subadult ablated lines were mostly local - 85.3% were local within 5 km and 100.0% within 30 km (Fig. 5)
- Variations in mobility profiles required additional categories, so five mobility categories were determined (see Fig. 6)
 - Nearly ½ of the individuals had non-local values, originating beyond 5km but only 20% of the values were non-local within 30 km (Fig. 6)
- Comparative Early Christian site analysis (Fig.1)
 - Alvastra: Both sites have a large cemetery, potentially served as magnet for Christian burial or affiliation¹
 - Sigtuna: Varnhem had more non-local values and greater genetic diversity despite no commercial center²

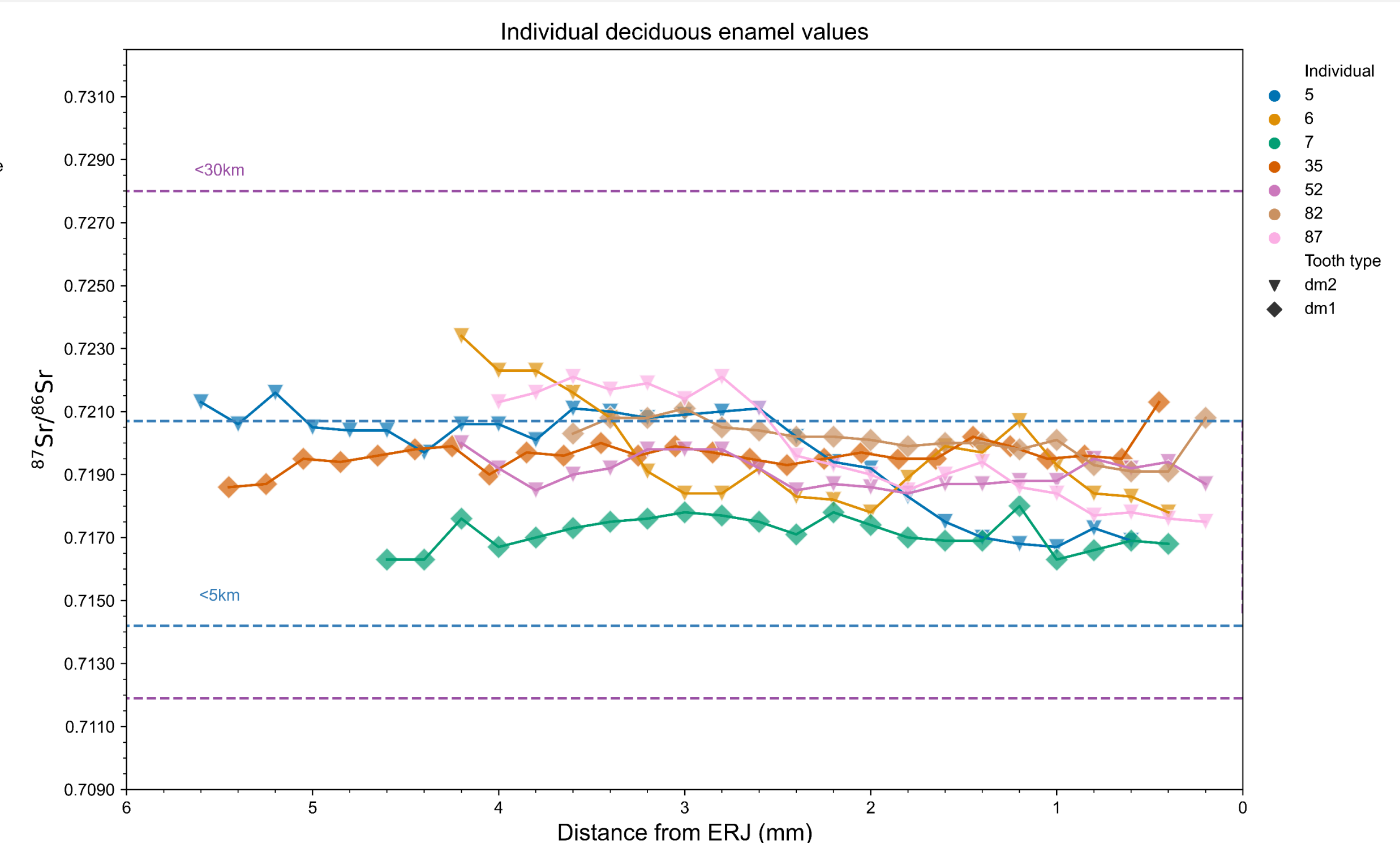
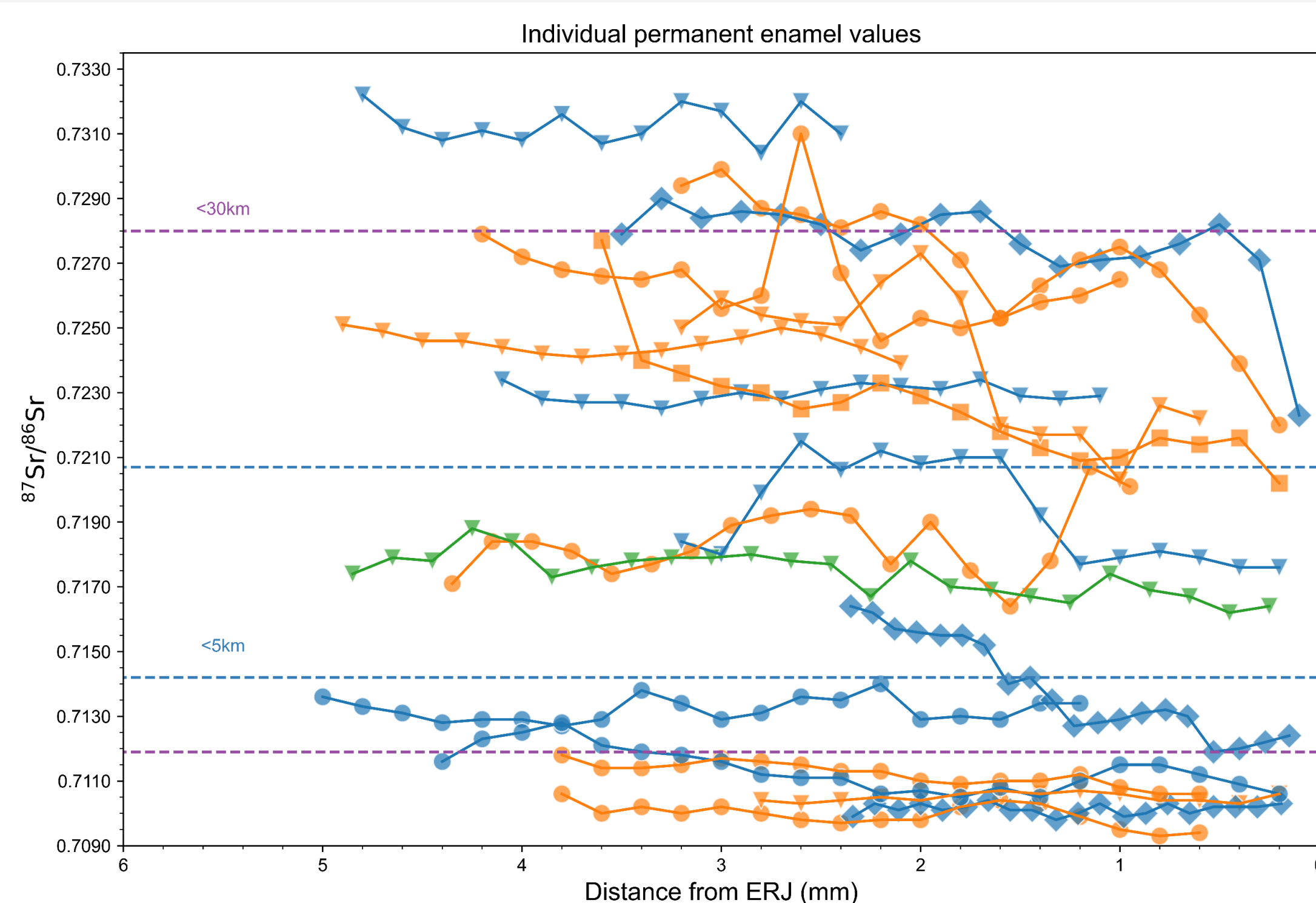
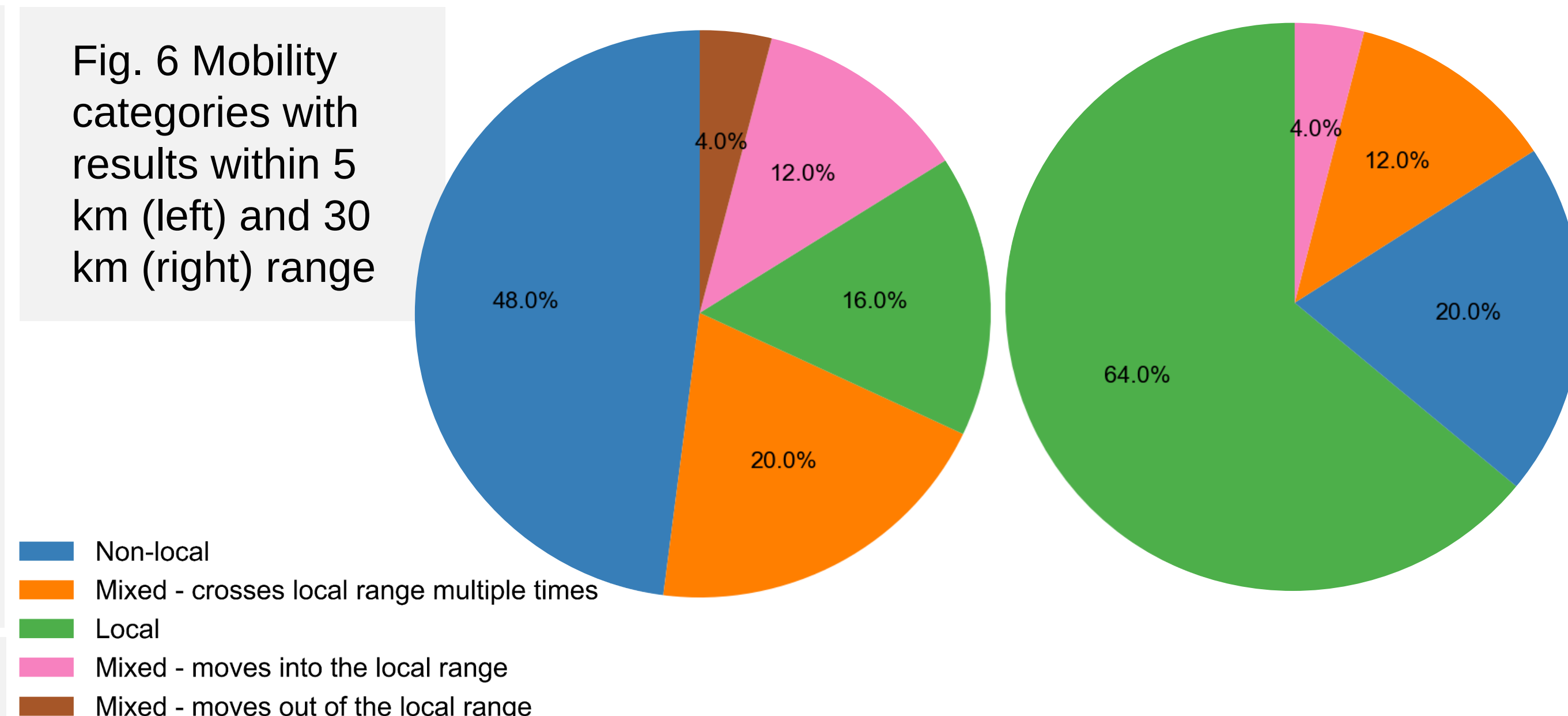


Fig. 5 ⁸⁷Sr/⁸⁶Sr values from permanent (left) and deciduous (right) molars plotted against the distance from the enamel-root junction (ERJ)

Conclusion

- High adult mobility, contrasted with low subadult mobility, suggests that Varnhem attracted incoming families
- Evidence indicates that Varnhem functioned as a regional Christian hub, drawing individuals from within 30 km and beyond
- These patterns highlight Varnhem's central role in the social and religious transformation of the region

Fig. 6 Mobility categories with results within 5 km (left) and 30 km (right) range



References

1. Vretemark, M. and Axelsson, T., 2008. The Varnhem Archaeological Research project: A new insight into the Christianization of Västergötland. Viking. Mediev. Scand. 4, 209–219.
2. Krzewińska, M., et al., 2018. Genomic and Strontium Isotope Variation Reveal Immigration Patterns in a Viking Age Town. Curr. Bio., 28, 2730-2738.e10.

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